Waveguides and Antennas [EE3140]

Programming Assignment - 1

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1 Part

2 Part

In this question we are supposed to design a L-type matching networks for the following complex loads. Given $Z_0 = 75\Omega$ and $f_0 = 1$ Ghz.

2.A $Z_L = (150 + j45)\Omega$

To solve:

$$z_l = \frac{150 + j45}{75} \Longrightarrow 2 + j\frac{3}{5} = 2 + j0.6 \quad \text{(Normalisation)}$$

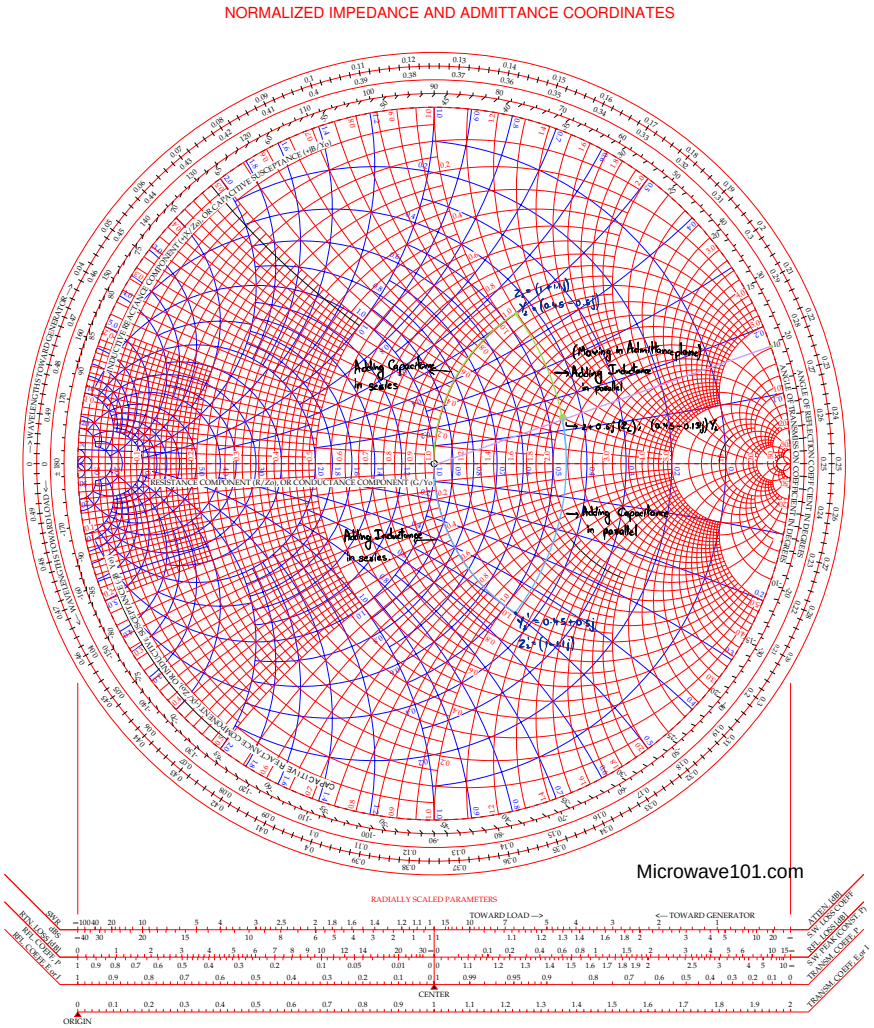


Figure 1: The work on ZY smith chart to design the L network

As evident in the annotated smith chart, we are going to need a **reactive element** in parallel first, and then an additional element in series. The 2 possible solutions are the following, annotated with green and blue sketches in the above chart.

On the Green line annotated solution, we need:

Movement on Admittance Circle

$$j\omega L_{\text{Parallel}} = -\frac{1}{0.37j} \times Z_0 \Longrightarrow j202\Omega$$

$$L_{\text{Parallel}} = \frac{202}{2\pi \times 10^9} \Longrightarrow 32.14nH$$

Movement on Impedance Circle

$$\frac{1}{j\omega C_{\text{Series}}} = -1.1j \times Z_0 \Longrightarrow -j82.5\Omega$$

$$C_{\text{Series}} = \frac{1}{2\pi \times 10^9 \times 82.5} \Longrightarrow 1.92pF$$

On the Cyan/Blue annotated solution, we need:

Movement on Admittance Circle

$$\frac{1}{j\omega C_{\text{Parallel}}} = \frac{1}{0.63j} \times Z_0 \Longrightarrow \frac{119.04}{j}\Omega$$

$$C_{\text{Parallel}} = \frac{1}{2\pi \times 10^9 \times 119.04} \Longrightarrow 1.33pF$$

Movement on Impedance Circle

$$j\omega L_{\text{Series}} = 1.1j \times Z_0 \Longrightarrow 82.5j\Omega$$

$$L_{\text{Series}} = \frac{82.5}{2\pi \times 10^9} \Longrightarrow 13.13nH$$

2.B $Z_L = (30 - j15)\Omega$

To solve:

$$z_l = \frac{30 - j15}{75} \Longrightarrow 0.4 - j0.2 \quad \text{(Normalisation)}$$

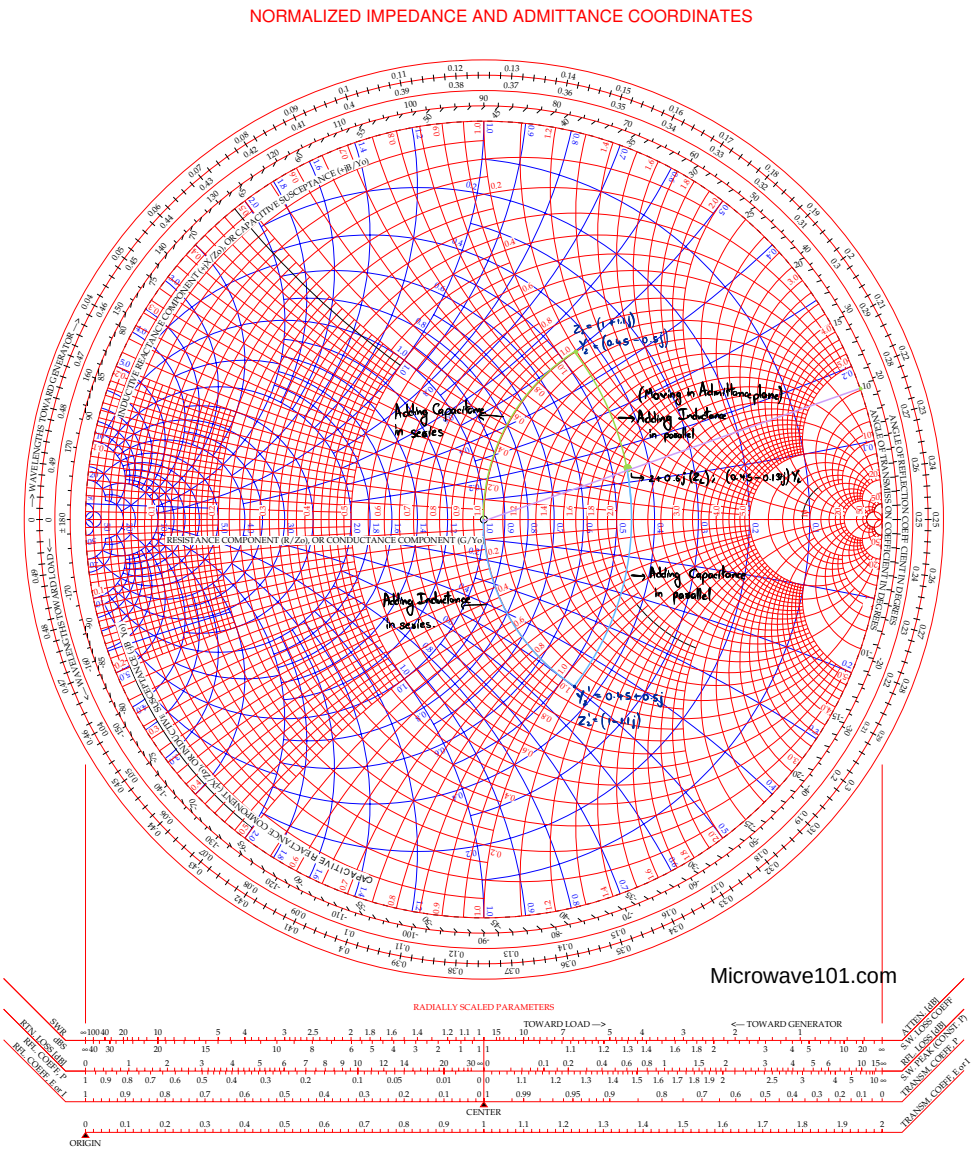


Figure 2: The work on ZY smith chart to design the L network

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Movement on Admittance Circle

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Movement on Impedance Circle

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$$C_{\text{Series}} = \frac{1}{2\pi \times 10^9 \times 82.5} \Longrightarrow 1.92pF$$

On the Cyan/Blue annotated solution, we need:

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Movement on Impedance Circle

$$j\omega L_{\text{Series}} = 1.1j \times Z_0 \Longrightarrow 82.5j\Omega$$

$$L_{\text{Series}} = \frac{82.5}{2\pi \times 10^9} \Longrightarrow 13.13nH$$

2.C $Z_L = (60 + j75)\Omega$

To solve:

$$z_l = \frac{30 - j15}{75} \Longrightarrow 0.4 - j0.2 \quad \text{(Normalisation)}$$

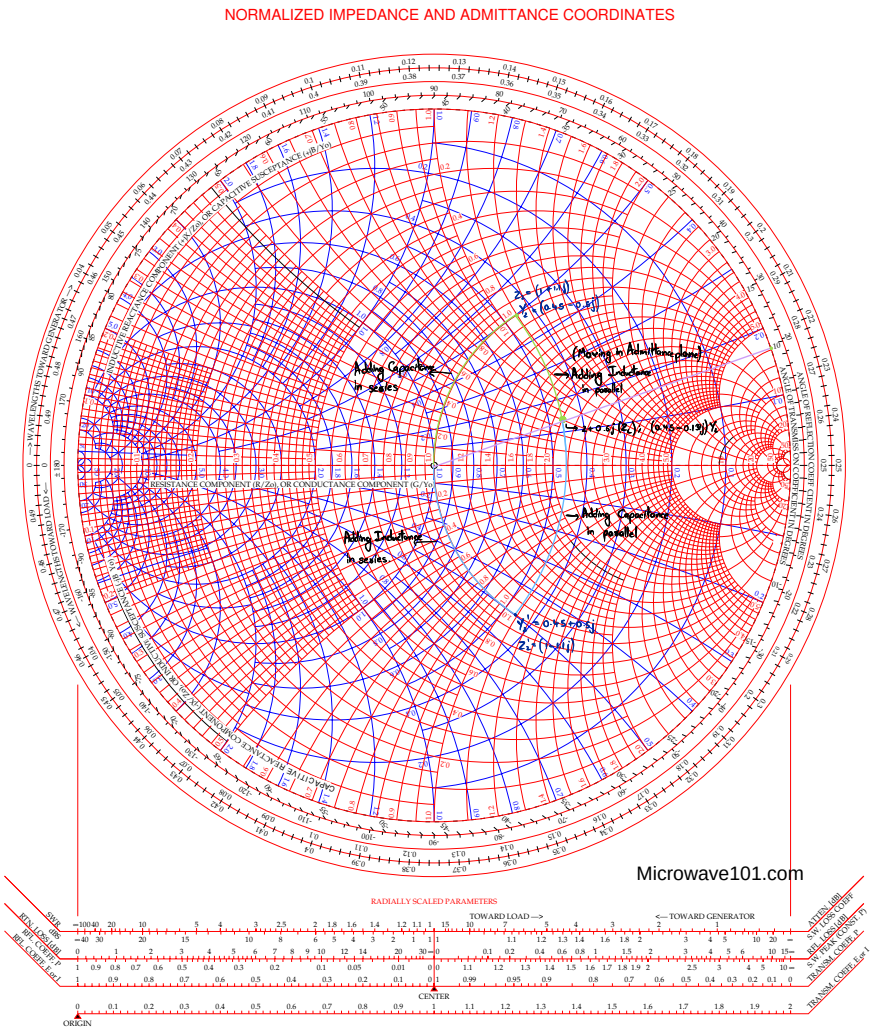


Figure 3: The work on ZY smith chart to design the L network

As evident in the annotated smith chart, we are going to need a **reactive element** in parallel first, and then an additional element in series. The 2 possible solutions are the following, annotated with green and blue sketches in the above chart.

On the Green line annotated solution, we need:

Movement on Admittance Circle

$$j\omega L_{\text{Parallel}} = -\frac{1}{0.37j} \times Z_0 \Longrightarrow j202\Omega$$

$$L_{\text{Parallel}} = \frac{202}{2\pi \times 10^9} \Longrightarrow 32.14nH$$

Movement on Impedance Circle

$$\frac{1}{j\omega C_{\text{Series}}} = -1.1j \times Z_0 \Longrightarrow -j82.5\Omega$$

$$C_{\text{Series}} = \frac{1}{2\pi \times 10^9 \times 82.5} \Longrightarrow 1.92pF$$

Movement on Admittance Circle

$$\frac{1}{j\omega C_{\text{Parallel}}} = \frac{1}{0.63j} \times Z_0 \Longrightarrow \frac{119.04}{j}\Omega$$

$$C_{\text{Parallel}} = \frac{1}{2\pi \times 10^9 \times 119.04} \Longrightarrow 1.33pF$$

Movement on Impedance Circle

$$j\omega L_{\text{Series}} = 1.1j \times Z_0 \Longrightarrow 82.5j\Omega$$

$$L_{\text{Series}} = \frac{82.5}{2\pi \times 10^9} \Longrightarrow 13.13nH$$

On the Cyan/Blue annotated solution, we need:

3 Part

3.A

As discussed in the class, if we assume that ω is high enough to satisfy the conditions: $R \ll \omega L$ and $G \ll \omega C$, then we can consider that transmission line, a ***low loss*** transmission line case, and the following quantities can be calculated by:

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \approx \sqrt{\frac{L}{C}} \Longrightarrow Z_0 = \sqrt{\frac{250 \times 10^{-9}}{100 \times 10^{-12}}} = 50\Omega$$

$$v_p = \frac{1}{\sqrt{LC}} \Longrightarrow v_p = \frac{1}{\sqrt{25 \times 10^{-18}}} = 2 \times 10^8 ms^{-1}$$

The above phase/propagation velocity is physical and valid, since it's lower than the intrinsic velocity of EM waves in Free Space / Vacuum ($c_0 \approx 3 \times 10^8 ms^{-1}$).

3.B

To determine the values of Reflection co-efficient Γ at the given 2 points, we:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma_{\text{At Load}} = \frac{100 - 50}{100 + 50} \Longrightarrow \frac{1}{3}$$

In the Switch is closed condition

$$\Gamma_{\text{At Source}} = \frac{25 - 50}{25 + 50} \Longrightarrow -\frac{1}{3}$$

In the Switch is open condition

$$\Gamma_{\text{At Source}} = \frac{\infty - 50}{\infty + 50} \Longrightarrow 1$$

Some other important quantities that we are going to need are:

$$T_{\text{Single Way}} = \frac{d}{v_p} \Rightarrow \frac{20}{2 \times 10^8} = 100ns$$

$$V_{\text{Initial}} = 1 \times \frac{Z_0}{Z_0 + R_s} \Rightarrow \frac{50}{75} = \frac{2}{3}V$$

Also according to the question, we have the switch closed at $t = 0s$ (Inferred it's was open before this), and then opens again at $t = 250ns$.

In the case where we were calculating the Γ at the load and source, we are essentially seeing an impedance discontinuity, and had to satisfy the general voltage and current equations $V(z) = V_0^+ \exp(-j\beta z) + V_0^- \exp(j\beta z)$ and $I(z) = I_0^+ \exp(-j\beta z) - I_0^- \exp(j\beta z)$ continuity at the impedance mis-matched boundaries/z locations, along with satisfying $V(z) = Z(z)I(z)$. When the switch opens, we'll have the current at the location of switch to be zero. To calculate the extra voltage pulse we get when opening the switch, we solve the boundary condition in the following way.

$$I(0, 250ns) = \frac{2}{3Z_0} - \frac{2}{9Z_0} - \frac{2}{27Z_0} + \frac{V_2^+}{Z_0} = 0 \qquad \text{(The 2/9 Voltage pulse is moving in opposite direction)}$$

Then the Value of $V_2^+ = -\frac{10}{27}$

3.C Bounce Diagrams and Waveforms

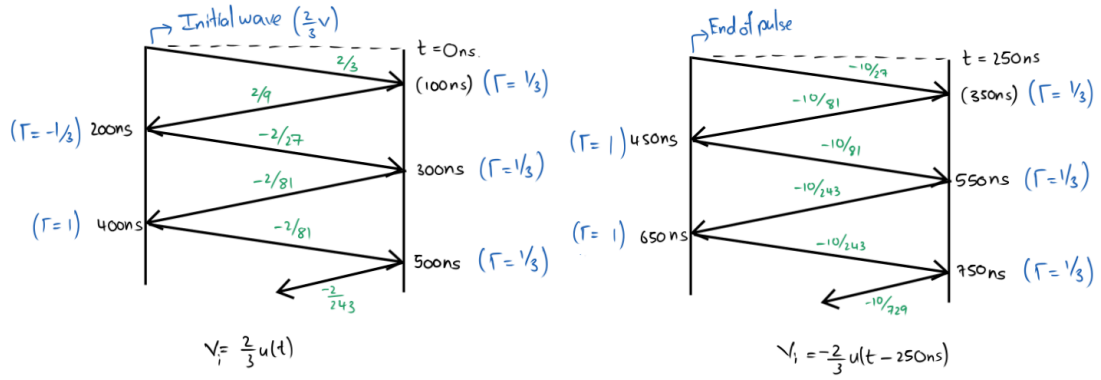


Figure 4: The bounds diagram of the given scenario

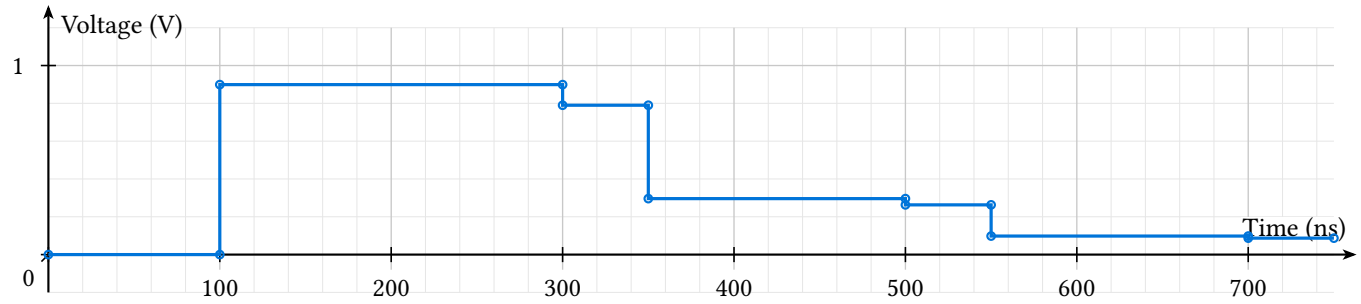


Figure 5: The $V(t)$ observed at the Load

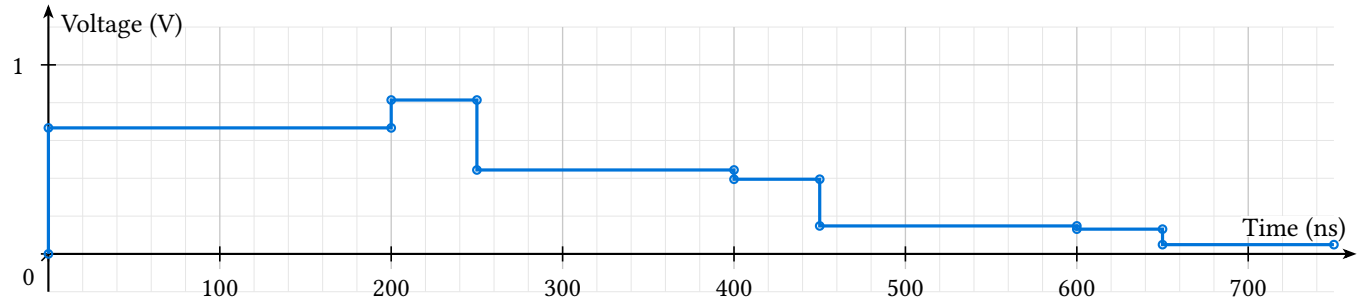


Figure 6: The $V(t)$ observed at the Source

The desmos versions of these graphs are available at, and we can also interactively view the voltages. However since we'll have infinite reflections in this case, they've been truncated to about 5 in the calculations.

3.D FTDT Simulation

For this part, I've hacked/modified my previous FTDT simulator used in the previous assignment. All the changes have been documented in the source code as comments.

3.D.A Lossless condition

The gist of this FTDT simulation is:

$$\frac{\partial V}{\partial z} = -L\frac{\partial I}{\partial t} - RI \qquad \text{(R is kept 0 in lossless condition)}$$

$$\frac{\partial I}{\partial z} = -C\frac{\partial V}{\partial t} - GV \qquad \text{(G is kept 0 in lossless condition)}$$

Which is solved by:

(Continued in the next page)

```
cCon = (dt / (grid->dx * Inductance));
for (int j = 0; j < grid->N - 1; j++) {
  grid->Hgrid[j] -= cCon * (grid->Egrid[j + 1] - grid->Egrid[j]);
}

cCon = (dt / (grid->dx * Capacitance));
for (int j = 0; j < grid->N; j++) {
  grid->Egrid[j] -= cCon * (grid->Hgrid[j] - grid->Hgrid[j - 1]);
}
```

,

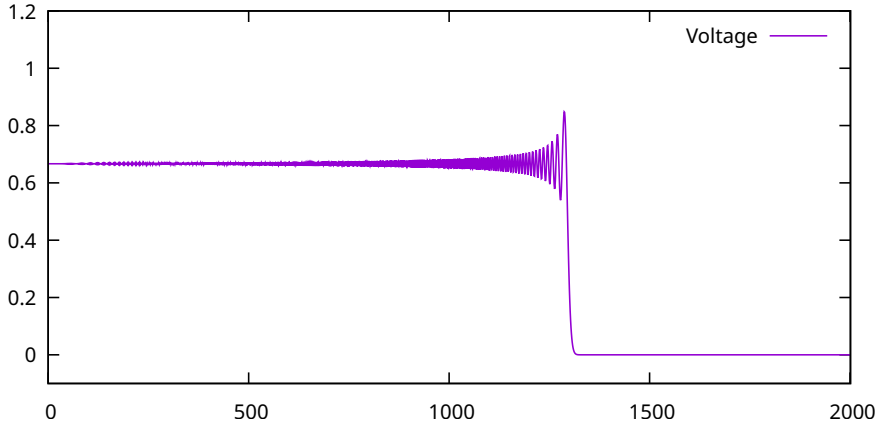


Figure 7: The Voltage across the transmission line at $65ns$, the x-axis being the domain mesh points (0-20m)

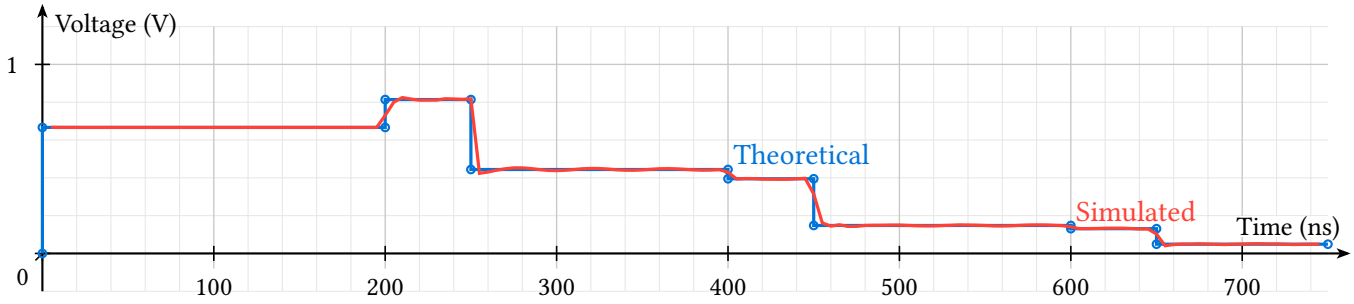


Figure 8: The $V(t)$ observed at the Source, Comparison between Theoretical and Simulated

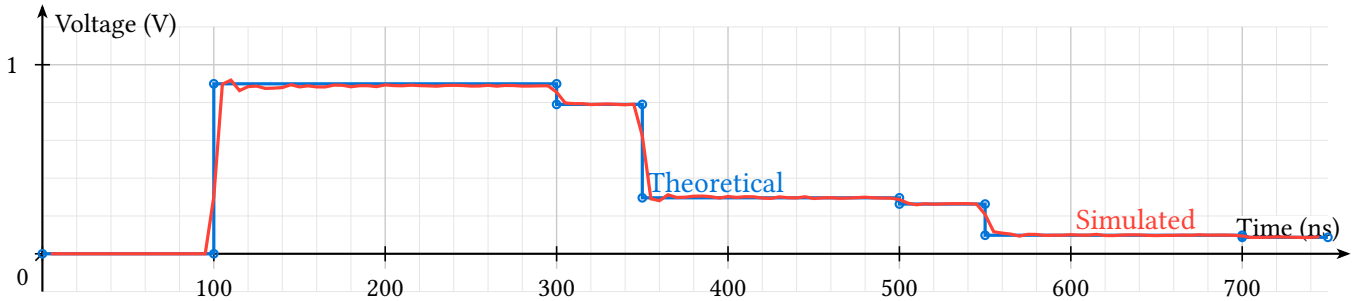


Figure 9: The $V(t)$ observed at the Load, Comparison between Theoretical and Simulated

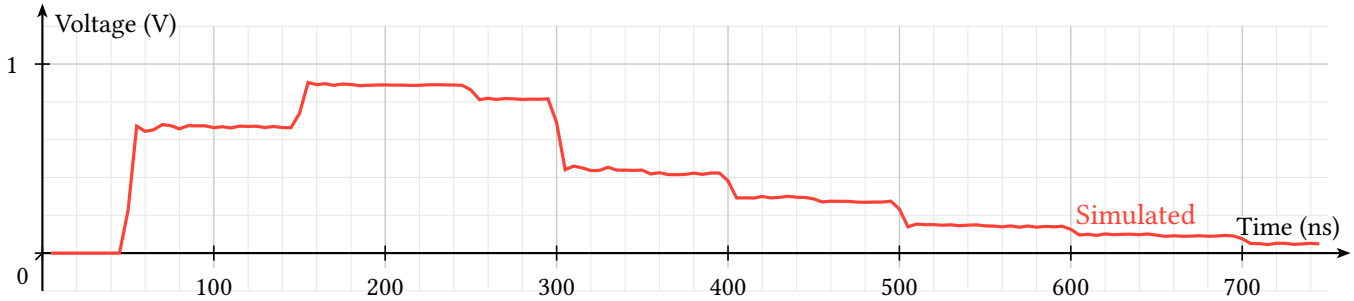


Figure 10: The $V(t)$ observed at the Mid point

3.D.B Lossy condition

```
cCon = (dt / (2.0 * Inductance));
double cCon1 = (1.0 - Resistance * cCon) / (1.0 + Resistance * cCon);
double cCon2 = (dt / (Inductance * grid->dx)) / (1.0 + Resistance * cCon);
for (int j = 0; j < grid->N - 1; j++) {
  grid->Hgrid[j] =
      cCon1 * grid->Hgrid[j] - cCon2 * (grid->Egrid[j + 1] - grid->Egrid[j]);
}

cCon = (dt / (2.0 * Capacitance));
cCon1 = (1.0 - Conductance * cCon) / (1.0 + Conductance * cCon);
cCon2 = (dt / (Capacitance * grid->dx)) / (1.0 + Conductance * cCon);
for (int j = 1; j < grid->N - 1; j++) {
  grid->Egrid[j] =
      cCon1 * grid->Egrid[j] - cCon2 * (grid->Hgrid[j] - grid->Hgrid[j - 1]);
}
```

,

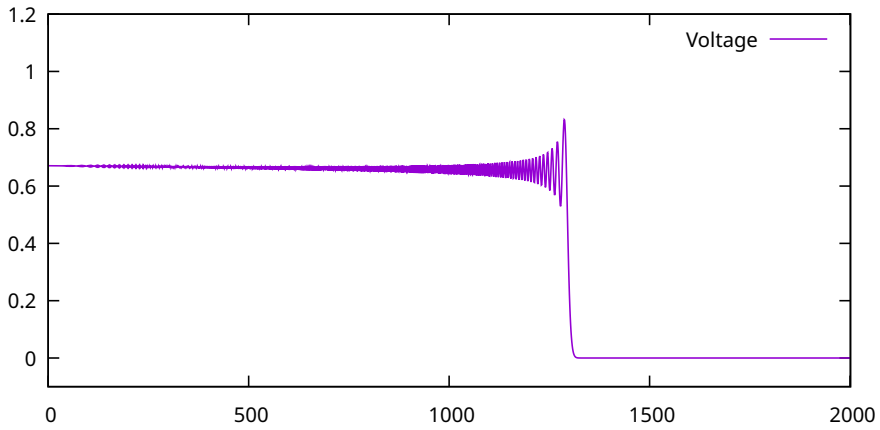


Figure 11: The Voltage across the transmission line at $65ns$, the x-axis being the domain mesh points (0-20m)

The Theoretical data in the following plots is still the lossless case, and is just there for Comparison. And as expected the peak/max voltages that we see in the simulation are lower than expected, while keeping the exact same shape. This essentially means the transmission line is still dispersion-less.

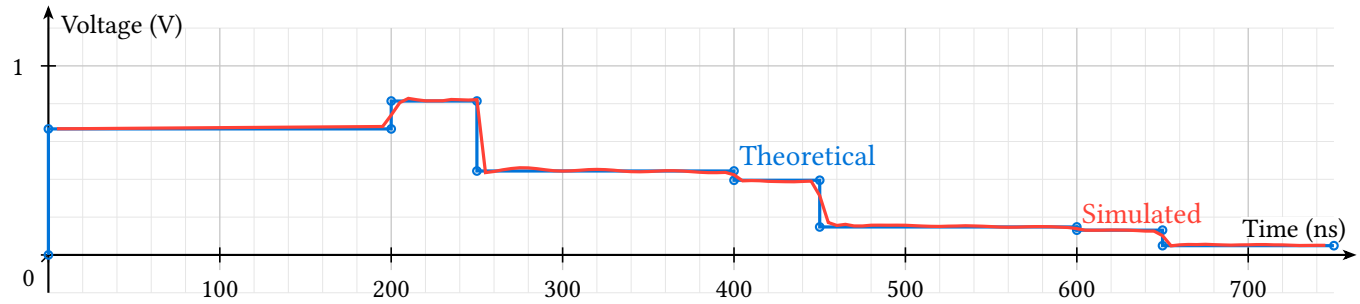


Figure 12: The $V(t)$ observed at the Source, Comparison between Theoretical and Simulated

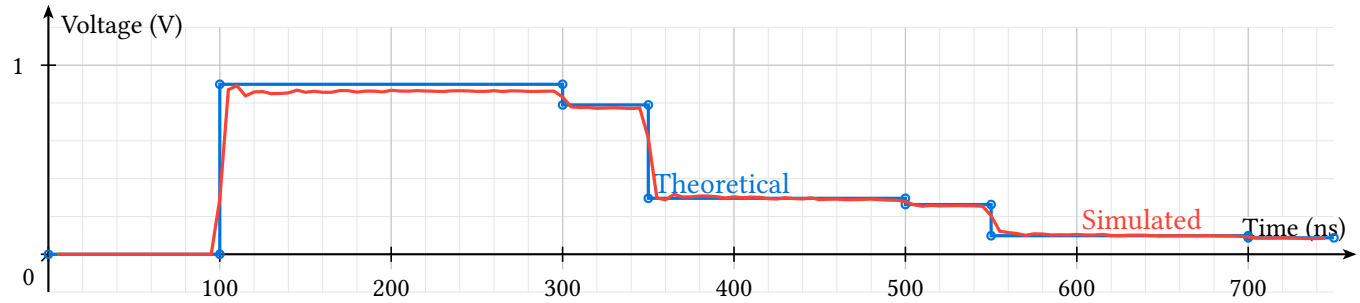


Figure 13: The $V(t)$ observed at the Load, Comparison between Theoretical and Simulated

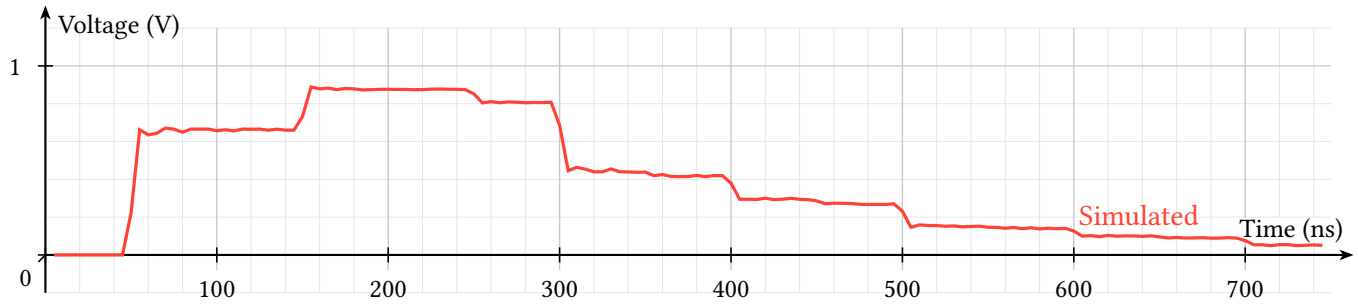


Figure 14: The $V(t)$ observed at the Mid point

3.D.C Dispersive - Dielectric Model

$$\varepsilon_r(\omega) = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + j\omega\tau}$$

$$C'(\omega) = C_{\infty'} + \frac{\Delta C'}{1 + j\omega\tau}, \quad \Delta C' = C_{\infty'}\left(\frac{\varepsilon_s - \varepsilon_\infty}{\varepsilon_\infty}\right)$$

$$\tau\frac{\partial P'}{\partial t} + P'(t) = \Delta C' V(t)$$

$$\frac{\partial V}{\partial z} = -L'\frac{\partial I}{\partial t} - R'I$$

$$\frac{\partial I}{\partial z} = -C_{\infty'}\frac{\partial V}{\partial t} - \frac{\partial P'}{\partial t} - G'V$$

$$v_{p,\infty} = \frac{1}{\sqrt{L'C_{\infty'}}}, \quad v_{p,0} = \frac{1}{\sqrt{L'(C_{\infty'} + \Delta C')}}$$

Which is solved by:

```
const double dC_p = Capacitance * (eps_s - eps_inf) / eps_inf;
const double alpha = (2.0 * tau - dt) / (2.0 * tau + dt);
const double beta_p = (2.0 * dt * dC_p) / (2.0 * tau + dt);
const double cB = beta_p / (2.0 * dt);
const double cP = (1.0 - alpha) / dt;

cCon = (dt / (2.0 * Inductance));
double cCon1 = (1.0 - Resistance * cCon) / (1.0 + Resistance * cCon);
double cCon2 = (dt / (Inductance * grid->dx)) / (1.0 + Resistance * cCon);
for (int j = 0; j < grid->N - 1; j++) {
  grid->Hgrid[j] =
      cCon1 * grid->Hgrid[j] - cCon2 * (grid->Egrid[j + 1] - grid->Egrid[j]);
}

memcpy(grid->Hgrid_prev, grid->Egrid_prev, sizeof(double) * grid->N);

for (int j = 1; j < grid->N - 1; j++) {
  double dI = (grid->Hgrid[j] - grid->Hgrid[j - 1]) / grid->dx;
  double V_old = grid->Egrid[j];
  double P_old = grid->Hgrid_prev[j];

  double V_new = (V_old * (Capacitance / dt - cB - Conductance / 2.0) - dI +
                  cP * P_old) /
                  (Capacitance / dt + cB + Conductance / 2.0);

  grid->Egrid_prev[j] = alpha * P_old + beta_p * (V_new + V_old) / 2.0;
  grid->Egrid[j] = V_new;
}
```

Where I'm using grid array structures grid->Egrid_prev, and grid->Hgrid_prev as P(Polarisation) and P_old arrays

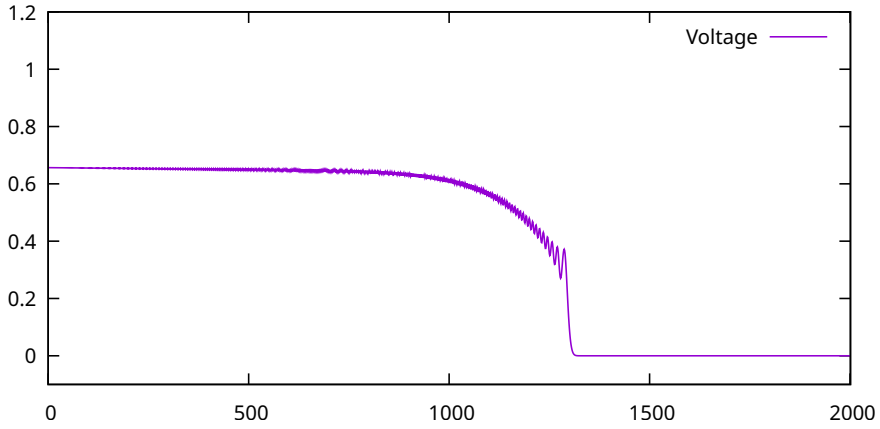


Figure 15: The Voltage across the transmission line at $65ns$, the x-axis being the domain mesh points (0-20m)

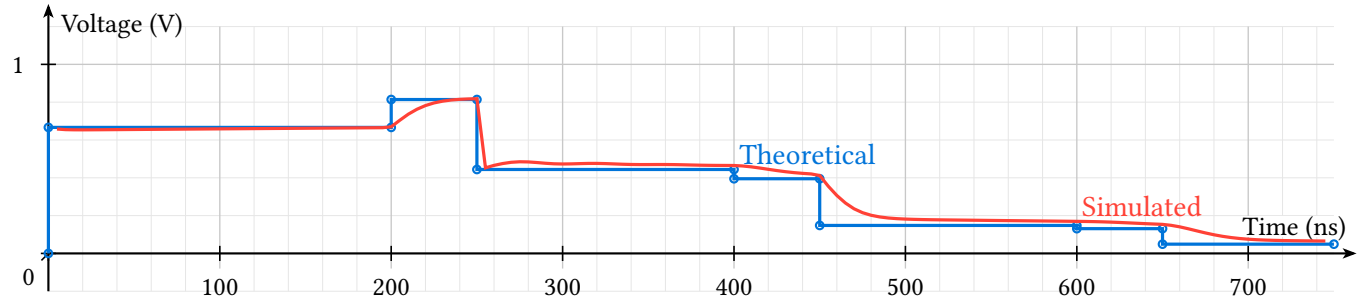


Figure 16: The $V(t)$ observed at the Source, Comparison between Theoretical and Simulated

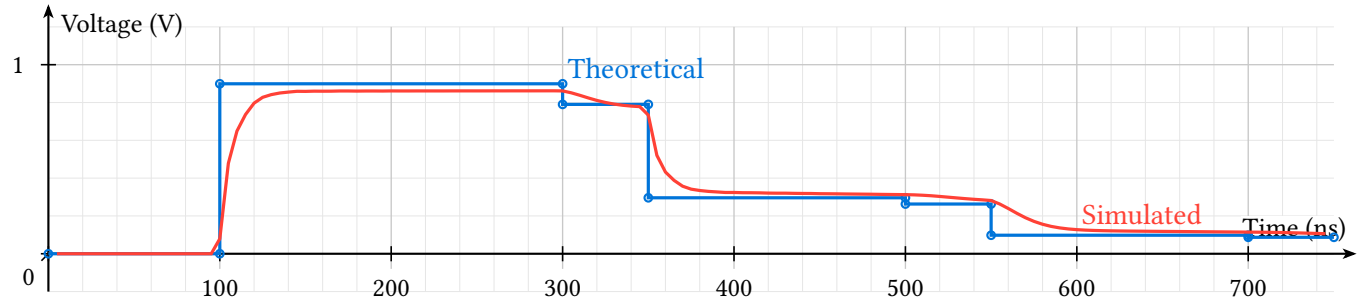


Figure 17: The $V(t)$ observed at the Load, Comparison between Theoretical and Simulated

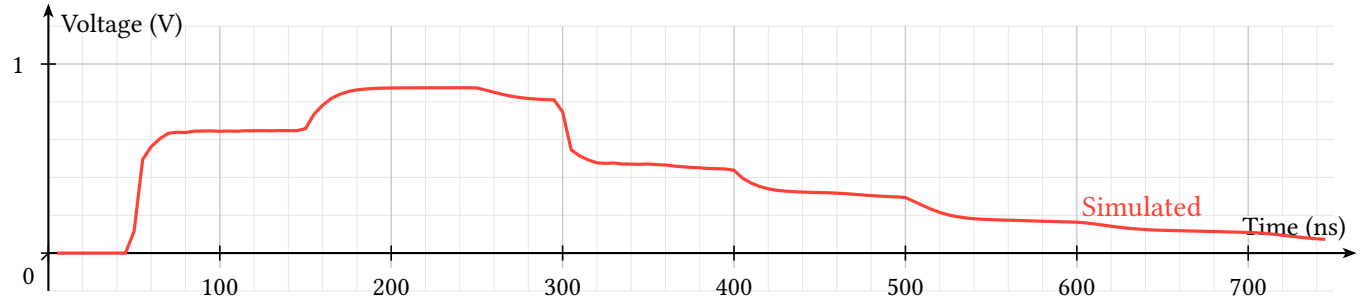


Figure 18: The $V(t)$ observed at the Mid point

And as we can see the output from the dielectric dispersion model is much more smeared, since the phase velocities for different frequencies aren't the same anymore, like in the case of Lossless or Low-Loss case.

3.D.D Comparison between the Theoretical/Analytical and Simulated results

As we can see from the above graphs, there's **severe ringing visible in the output Voltage and Current** which is the result of us **not having a smooth excitation source**, which will not contain the infinite series of harmonics that can't exactly be represented in the finite discrete grid we have, and hence we can't expect the sharp rise and fall times in the voltage. A way to reduce the effect of this would be to use more finer meshing but that is prohibitively expensive as we'd need much more compute time, and reality also won't let us have the ideal theoretical response we are hoping for.

On the plus side, we see that there's a **very nice match between the expected amplitudes of Lossless Simulation and Analytical data**. And for the lossy case, since the expected loss was very low, we weren't able to observe much exponential decay, but the amplitude compared the lossless case was lower, while keeping the shape intact (So basically dispersion less), which was expected.

GitHub Repository

The github repository can be accessed here. It will include a readme for this Programming assignment holding all the animated Gifs, and the requested source code.

AI Disclosure

There was use of Generative AI (ChatGPT) to solve the Questions - 1 (Understanding the FEM and writing code for it, in python), and Question - 2 (For the smith Chart diagrams and the plots) and Question - 3 (Idea to solve the Dielectric Model part), Code completely written by hand for the 3rd Question. All the use was related to coding and understanding the concept and problems, the report is hand written.

END